Week 14 Milestone Report: Testing and Release Readiness

Course: CS3320 Database Systems, Instructor Dr. Chibuzo Ukegbu
Student: Shritej Gunda (individual project)
Project: Factory Sherlock: Manufacturing Simulation, Quality Control, and Defect Investigation on MySQL
Date: [submission date]
Repository: https://github.com/shritej6/factory-sherlock
Commit or tag: [commit hash after push]. The release candidate tag is planned as v1.0.0-rc1 on the commit that merges feature/completion-integration.

1. Completion status, incomplete items, and blockers

The Week 14 row of docs/milestones/README.md lists eight required artifacts; the walkthrough video from the media plan is the ninth item. Status on the integration branch (head c978744) on 2026-11-07:

Item	Status	Evidence
Release candidate	Complete on the branch, tag pending	feature/completion-integration holds every feature and fix merge since the first push on 2026-11-04; five GitHub Actions workflows; all local gates pass (section 2). Tag v1.0.0-rc1 is [to be created after the merge to main].
Completed test matrix	Complete	docs/testing/test-matrix.md (TM-01 to TM-34, one row per behaviour and one per Playwright spec) and testing-strategy.md were rewritten on 2026-11-07 from the suite runs in section 2; every cited test name was checked against backend/tests and frontend/src.
Defect and fix evidence	Complete	Nine review and fix reports from the final wave, condensed in section 2; every named test exists in the worktree.
Final ERD and schema	Complete	docs/media/diagrams/er-diagram.svg; database/schema/schema.sql regenerated in c978744; three migrations with downgrades.
Tested README	Complete	README commands executed in this worktree (section 2); the stale statements found during that run were corrected the same day (commit 9e7cd7a).
Demo data	Complete	make reset-demo against database fs_w08 on 2026-11-07; counts in section 2.
Report draft	Complete, student sections pending	This document plus docs/milestones/week12-milestone-report.md and week13-milestone-report.md; the bracketed prompts in sections 4, 6, 8 and 9 of each are for the student.
AI evidence package	Complete, student annotations pending	docs/ai_evidence: README, ai-use-record.md (50 rows), seven conversation records, reflection-notes.md.
Walkthrough video and screenshot refresh	In progress	docs/media/videos/README.md lists six videos, all Pending; the nine screenshots are dated 2026-11-03 and predate pagination, the control chart, the import panel, and the overdue check. No new files exist in this worktree.

Blockers: none that stop the submission. Open risks are the recording time for the video and the first GitHub Actions runs of the Compose Smoke and End-to-End workflows, which happen only when the branch is pushed. [Add any blocker you see, or write "none".]

2. Required technical artifacts and repository commit or tag

Release candidate

Branch feature/completion-integration, head c978744 (chore: regenerate schema and OpenAPI exports). The merge commits name each feature and the six fix merges of the final wave (0ffdf3e, ae949eb, 758b4e6, 5fcdebf, 9325d87, 582c360).

Workflow	File	Checks
Backend CI	backend-ci.yml	Against a mysql:8.4 service: init_databases.sh, alembic upgrade head, ruff check, ruff format --check, mypy app, pytest with coverage
Frontend CI	frontend-ci.yml	pnpm install --frozen-lockfile, eslint, prettier check, tsc on three tsconfigs, vitest, vite build
Docs Lint	docs-lint.yml	Fails on em dashes, en dashes, or emoji in Markdown; renders every .mmd diagram with mermaid-cli (allowed to fail)
Compose Smoke	compose-smoke.yml	Starts mysql from docker-compose.yml, checks both schemas and the least-privilege grants as root, connects as the application user over TCP, confirms mysql.user is denied
End-to-End	e2e.yml	Migrates and seeds a mysql:8.4 service with the demo scenario, installs Chromium, runs the Playwright suite (backend port 8013, Vite port 5183)

Every gate runs locally from the Makefile: make lint (ruff check, ruff format --check, mypy, eslint, tsc), make test-backend (pytest against the <MYSQL_DATABASE>_test schema), make test-frontend, make check-docs, make migrate, make reset-demo, make export-schema, make export-openapi; the browser suite is pnpm test:e2e in frontend/. Observed in this worktree on 2026-11-07: make lint printed "All checks passed!", "122 files already formatted", and "Success: no issues found in 80 source files", with eslint and tsc clean; make check-docs passed; make test-backend against fs_w08_test printed "481 passed, 1 warning in 222.18s". The Playwright suite passed 9 of 9 locally on 2026-11-07 at commit 11d5c3a (`pnpm test:e2e`, 4.9 s); the End-to-End workflow runs the same suite in CI.

Completed test matrix

docs/testing/test-matrix.md holds rows TM-01 to TM-34: the twelve required scenarios, one row per behaviour added since Week 10 (pagination, import, body caps, overdue alerts, the control chart, R7, status filters, list ordering indexes, the throttle, frontend accessibility), and one row per Playwright spec, each with Setup, Action, Expected result, Actual result, Status, and Evidence naming the test functions. docs/testing/testing-strategy.md describes the pyramid, the database isolation, the end-to-end layer, the five workflows, and the benchmark scripts. Both were rewritten on 2026-11-07 from these counts:

Layer	Count	Source
Backend unit (no database)	254 cases in 13 modules	uv run pytest --collect-only -q: 481 collected

Layer	Count	Source
Backend API (TestClient against MySQL)	174 cases in 16 modules	same collection
Backend database integration	53 cases in 6 modules	same collection
Frontend Vitest	118 tests in 25 files	pnpm -s test
Playwright end-to-end	9 tests in 5 spec files	analytics (1), investigation (1), login (2), roles (2), traceability (3)

About 53 percent of the backend cases are unit tests and 47 percent run against MySQL 8.4, so constraints, transactions, and the SQL files are exercised on the real server rather than mocked.

Defect and fix evidence

The final wave produced nine reports (fix reports F1 to F6, the accessibility pass, two documentation passes). Each row names the test that now guards the finding; every name was checked against the worktree.

#	Finding	Severity	Symptom	Fix	Guarding test
1	Shared address lockout and header spoofing	High	Five wrong passwords from one address locked every account behind the Vite proxy (all browsers arrive from 127.0.0.1); uvicorn trusted X-Forwarded-For from loopback	Account bucket checked first, address bucket only after a failure (401 becomes 429); uvicorn runs with --no-proxy-headers	test_failures_as_one_user_do_not_lock_another_user_from_the_same_address, test_missing_peer_address_is_reported_as_none (backend/tests/api/test_login_throttle_api.py)
2	Username normalisation	Medium	admin, ADMIN, and an accented spelling had separate buckets although the collation resolves them to one row	Key by user id when known, otherwise NFKD, mark-stripped, casefolded username	test_case_and_accent_variants_count_against_the_same_account (same file)
3	check/record race	Medium	Requests between check() and record_failure() could all pass the limit	check() reserves a pending slot under the lock	test_concurrent_checks_cannot_exceed_the_limit (20 threads, 5 admitted; backend/tests/unit/test_login_throttle.py)
4	CHECK constraint mapped to 503	Medium	MySQL errno 3819 arrives as OperationalError and was answered 503	classify_operational maps 3819 to 422 with the constraint name, 1406 to 422, connection errors to 503 "not reachable"	test_check_constraint_violation_from_the_database_is_a_422_with_the_constraint_name (backend/tests/api/test_error_contract.py); backend/tests/unit/test_error_mapping.py

#	Finding	Severity	Symptom	Fix	Guarding test
5	Per-row CSV writes	High	record_inspection per row: 10,507 statements, 1.47 s for 5,000 rows	In-memory evaluation, then chunked multi-row INSERTs in a savepoint: 15 statements, 0.11 s (scripts/development/benchmark_inspection_import.py)	test_valid_file_is_imported_with_the_same_rules_as_manual_entry, test_one_invalid_row_rolls_back_the_whole_file (backend/tests/api/test_inspection_import.py)
6	csv.Error 500	Medium	An oversized field or text after a closing quote crashed with 500	parse_csv catches csv.Error and answers 422 with field "file"	test_broken_csv_structure_is_a_file_error_not_a_crash (backend/tests/unit/test_inspection_csv.py)
7	Import race	Medium	A pair recorded by another request during the import hit uq_inspection_unit_spec without a row-level message	errno 1062 rolls back to the savepoint, re-reads with a locking read, reports the rows as 422; other integrity errors re-raised	test_pair_recorded_by_another_request_during_the_import_is_a_row_error, test_other_integrity_errors_are_not_mistaken_for_the_race (test_inspection_import.py)
8	Decimal parser	Low	Decimal() accepted 1_000, 2.004e1, and non-ASCII digits	Fullmatch on sign, ASCII digits, at most three decimals before Decimal()	test_bad_measured_values_name_the_row_and_field (parametrised, test_inspection_csv.py)
9	as_of forgery	Medium	An operator could post {"as_of": ...} to raise alerts early or back-date created_at	The endpoint takes no body and uses the server clock; only the CLI job keeps --as-of	test_client_cannot_choose_the_check_date (backend/tests/api/test_alert_checks_api.py)
10	Duplicate overdue alerts	Medium	Two overlapping checks could both find no open alert and insert two	SELECT ... FOR UPDATE on the orders and their open alerts	test_concurrent_checks_raise_one_alert_per_order (backend/tests/integration/test_alert_order_link.py)
11	Job failure handling	Medium	A failure in the job left a bare traceback, no structured log line, no defined exit status	run_check rolls back; main logs under operation=check_overdue_orders and returns 1	test_job_rolls_back_logs_and_exits_one_on_failure (same file)
12	Control limits from average sample size	High	Query 10 divided the pooled standard deviation by SQRT(avg_sample_size), so every subgroup got the same limits	Per-row SQRT(sample_count) in database/queries/10_characteristic_control_chart.sql; the chart draws stepped limits	test_control_chart_limits_widen_for_smaller_subgroups (backend/tests/integration/test_queries.py); "judges each subgroup against its own limits" (frontend/src/utils/chartTransforms.test.ts)

#	Finding	Severity	Symptom	Fix	Guarding test
13	Missing list ordering indexes	Medium	Four paginated lists sorted with a filesort over the whole table; query 10 had no index covering both predicates	Migration 6c1f0d2a9b47 adds five indexes	test_list_ordering_indexes_exist_with_time_then_id, test_composite_index_replaces_the_implicit_specification_fk_index (backend/tests/integration/test_list_ordering_indexes.py)
14	Six-request order count fan-out	Medium	The overview issued one list request per order status to count active orders	GET /production-orders/status-counts from one GROUP BY	test_order_status_counts_match_the_list_totals (backend/tests/api/test_status_filters_api.py); "counts active orders from one status-counts request" (frontend/src/features/overview/OverviewPage.test.tsx)
15	Client-side filtering over 500 rows	Medium	Batch status filter and open investigation count fetched 500 rows and filtered in the browser	Repeatable status filter in SQL before pagination, so total reflects the filter; order_number joined with selectinload	test_batches_status_total_reflects_filter_before_paging, test_batch_list_loads_order_numbers_without_a_query_per_row (test_status_filters_api.py)
16	Bare-array tests broken by the pagination envelope	Low	Eight API tests and the e2e helper still read a bare array after six lists moved to {items, total, limit, offset}	Tests read .json() ["items"]; the e2e helper requests limit 500 (7bca7ac, a95df14)	test_inspection_import.py and test_alert_checks_api.py pass again; frontend/e2e/investigation.spec.ts
17	Accessibility contrast and focus	Medium	ink-faint text at 4.42:1 and the request id line at 3.95:1; closing an inline form dropped focus to body; the drawer opened without moving focus	Token shades at 4.5:1 or better, focus returns to the toggle, drawer focuses its first link	"returns focus to the toggle when the form closes itself" (frontend/src/components/ui/Disclosure.test.tsx); "moves focus into the drawer when it opens and back to the menu button on Escape" (frontend/src/app/AppLayout.test.tsx)
18	Pagination showing an empty range	Low	When the offset ran past the total the table showed an empty page	Pagination clamps to the last page with rows and emits the corrected offset	"clamps back to the last page with rows when the offset runs past the total" (frontend/src/components/ui/Pagination.test.tsx)

Two corrections landed after the reports: catalog_service converts only errno 1062 to 409 (6635d8a, backend/tests/unit/test_catalog_conflict.py), and the frontend no longer sends a body to the overdue check (1140e0e, "posts without a body" in OverviewPage.test.tsx).

Final ERD and schema

database/schema/schema.sql was regenerated from the migrated database in c978744: 17 application tables plus alembic_version. Compared with Week 10 it carries alerts.production_order_id with fk_alerts_production_order (ON DELETE SET NULL) and ix_alerts_production_order, and the five indexes from revision 6c1f0d2a9b47: ix_inspections_time_id (inspected_at, id), ix_inspections_spec_time (specification_id, inspected_at), ix_production_batches_started_id (started_at, id), ix_production_orders_created_id (created_at, id), ix_investigations_created_id (created_at, id). The implicit specification_id index on inspections is gone: MySQL drops it when a composite index with the same leading column is created, the upgrade drops it explicitly when MySQL kept it, and the downgrade recreates it before dropping the composite index (error 1553 otherwise). The three migrations in backend/migrations/versions each have a downgrade: 3a5e982c3679 (tables), 383273c26ef7 (alerts.production_order_id), 6c1f0d2a9b47 (indexes). docs/database/database-design.md section 6 records the index rationale with EXPLAIN observations and the lock class of each migration operation. docs/media/diagrams/er-diagram.svg is rendered from er-diagram.mmd, which names production_order_id on alerts.

Tested README

README.md commands executed in this worktree on 2026-11-07 against the shared local MySQL 8.4.0 server with .env pointing at fs_w08: cd backend && uv sync; uv run alembic upgrade head (383273c26ef7 and 6c1f0d2a9b47 applied on top of 3a5e982c3679); cd frontend && pnpm install --frozen-lockfile; make reset-demo (dropped 18 tables, re-ran the three migrations, seeded the scenario); make lint; make test-frontend; make test-backend; make check-docs; make help. make db-init and make db-start were not run because the shared server was already up; docker compose up was not run because Docker is not installed on this machine (the Compose Smoke workflow covers that path).

That run found README.md statements that no longer matched the code: "Six evidence rules (R1 to R6)" (there are seven), "Twelve SQL files" (there are 14), "31 paths" (openapi.json has 35 paths and 44 operations), the 2026-11-03 test table, "eight records" (nine ADRs), "the compose file has not been executed", "no rate limiting or account lockout", "without pagination", and roadmap items 2 to 6 still listed as open. All of them were corrected in commit 9e7cd7a, which also updated CONTRIBUTING.md, CHANGELOG.md and the development roadmap.

Demo data

make reset-demo on fs_w08 (2026-11-07, 22:36 UTC) logged seed 42, orders_created 9, batches_created 35, units_created 700, inspections_created 1400, failed_inspections 51, events_created 12, alerts_created 2. Row counts queried afterwards with the mysql client:

Table	Rows
users (demo accounts)	4
products / product_specifications	3 / 6
machines	3 (CNC-01 IDLE, CNC-02 IDLE, CNC-03 WARNING)
defect_types / material_lots	5 / 9
production_orders / production_batches / production_units	9 / 35 / 700

Table	Rows
batch_materials	47
inspections (FAIL)	1,400 (51)
inspection_defects	51
machine_events	12
alerts	2 (MACHINE_FAULT CRITICAL on CNC-02, HIGH_DEFECT_RATE WARNING on CNC-03)
investigations / investigation_evidence / investigation_batches	0 / 0 / 0

The seed opens no investigation; the case board screenshot and the end-to-end spec open one through POST /api/v1/investigations on CNC-03. alembic_version read 6c1f0d2a9b47 and @@sql_mode included ONLY_FULL_GROUP_BY.

Report draft

This document is the Week 14 draft; the Week 12 and Week 13 reports sit beside it in docs/milestones. All three build to DOCX and PDF with docs/milestones/build_documents.py, which also counts the bracketed prompts that remain for the student.

AI evidence package

docs/ai_evidence holds README.md (tools, verification approach, honesty rules), ai-use-record.md (43 numbered rows dated 11-03 to 11-07), six conversation records (schema and normalization; simulation state machine revision; rejected Alembic downgrade; ORM aggregation bug; environment and client defaults; parallel feature integration), and reflection-notes.md with the prompts for the Week 8 and Week 15 reflections. The student annotation at the end of each conversation record is still bracketed.

3. Decisions, alternatives, assumptions, constraints, and rationale

Decision	Alternatives considered	Rationale
Release scope: the functional unique index on open ORDER_OVERDUE alerts was probed and not added	CREATE UNIQUE INDEX over CASE WHEN alert_type = 'ORDER_OVERDUE' AND acknowledged_at IS NULL THEN production_order_id END; MySQL 8.4.0 accepted it on a scratch database and rejected a second open alert with error 1062	The locking read in alert_service already serialises concurrent checks and is covered by a two-session test; the index would also require the service to treat 1062 on that key as "already raised". Recorded as a follow-up
Videos hosted externally, not in Git LFS	Git LFS for MP4 files	docs/media/media-plan.md keeps large media out of the repository and links recordings from docs/media/videos/README.md; LFS would add a dependency to every clone

Decision	Alternatives considered	Rationale
No refresh tokens or revocation	Refresh endpoint plus a revocation list	SECURITY.md and ADR-0007 record the limitation: tokens expire after JWT_ACCESS_TOKEN_MINUTES, and get_current_user reloads the row so a deactivated user is rejected on the next request
MySQL default sql_mode written out in full, with ONLY_FULL_GROUP_BY	Leave the local server on its inherited mode, which had the flag off	CI runs mysql:8.4 with the default mode, so a query that only works with the flag off would pass locally and fail in CI. All 14 catalogue queries and the API and integration tests were run with the flag forced on before local_mysql.sh and docker-compose.yml were changed
The overdue check stays unscheduled	A cron entry or a scheduler thread in the API process	Scheduling is deployment configuration for a project that runs locally. The job is idempotent and reachable from make check-overdue and POST /analytics/alerts/checks/overdue-orders; docs/architecture/repository-structure.md states that nothing in the repository schedules it
Test database isolation by schema suffix and per-test TRUNCATE	Transaction rollback per test	backend/tests/conftest.py forces APP_ENV=test, which selects <MYSQL_DATABASE>_test, and refuses to run otherwise; migrations run once per session, every table is truncated before each test, reference data and demo users are re-applied. Rollback per test would hide the commit behaviour the API tests assert. Each parallel worktree used its own pair (fs_w01 to fs_w20 and their _test schemas)
The overdue endpoint ignores a body instead of rejecting it	Reject any body with 422	Rejecting would have broken the frontend until 1140e0e landed; FastAPI ignores an undeclared body

Assumptions and constraints: the demo scenario covers one production day on three machines; MySQL 8.4.0 is the only server version tested; the throttle state is per API process, so a multi-worker deployment would count failures per worker (a documented known discrepancy); timestamps are server local time without a zone.

4. AI help: what, why, accepted, revised, rejected, and verification

Tool: Claude Code (Anthropic). During the final wave it drafted the fixes, tests, and documentation changes on separate worktree branches, each gated by the full pytest suite against MySQL 8.4, ruff, mypy, ESLint, tsc, Vitest, and a production build before merge. Rows 29 to 43 of docs/ai_evidence/ai-use-record.md record the decisions, including two revisions (eight tests and the end-to-end helper rewritten for the pagination envelope; an unmigrated development database diagnosed from a 503 body) and one item left open (the 503 body for GET /health with MySQL stopped is still not asserted by an automated test).

[Write 100 to 150 words in your own voice: one suggestion from the final wave you accepted as drafted, one you changed and why, and one you rejected or deferred, each with the test, query, or EXPLAIN output you used to decide.]

5. Lessons learned at this milestone

- Parallel branches need a written contract before they start. The pagination envelope was decided on one branch while three sibling branches were written against bare arrays; eight API tests and the e2e helper failed at merge time (docs/ai_evidence/conversations/06-parallel-feature-integration.md).
- Run alembic upgrade head on every database after every merge. An unmigrated development database surfaced as "The database is not reachable" (503) because pymysql reports an unknown column (1054) as OperationalError; nothing in the code was wrong.
- MySQL drops the implicit index behind a foreign key when a composite index with the same leading column is created, but only when that index was auto-generated. Migration 6c1f0d2a9b47 handles both states so the upgrade is deterministic and the downgrade avoids error 1553.
- A local server can differ from CI in sql_mode. Writing the full MySQL 8.4 default into local_mysql.sh and docker-compose.yml removed the difference.
- A per-address login throttle behind a development proxy is a shared lock for everyone. The regression test uses two TestClient instances with different addresses so the behaviour is asserted, not assumed.
- [Add one lesson in your own words.]

6. Support or skill development needed for the next stage

[State any instructor support you need before the final presentation, for example a check of the Canvas submission format, or write "None at this time".]

[Name one skill you want to practice before the defense, for example reading EXPLAIN plans aloud or explaining the FOR UPDATE serialisation in alert_service.]

7. Next-step plan (final presentation and submission)

From the submission checklist in docs/milestones/README.md:

1. Merge feature/completion-integration to main, push, tag v1.0.0-rc1, and record the hash in the header of this report and of the Week 12 and 13 reports. [date]
2. Confirm the first GitHub Actions runs of all five workflows and record the run ids in docs/testing/test-matrix.md. [date]
3. Correct the stale README.md statements listed in section 2; rebuild the DOCX and PDF for Weeks 12, 13, and 14 with build_documents.py and check the page counts it prints. [date]
4. Replace every bracketed prompt in the three reports and in the six conversation records under docs/ai_evidence/conversations. [date]
5. Refresh the nine screenshots with scripts/development/capture_screenshots.py against make reset-demo; record the walkthrough video and add its link to docs/media/videos/README.md. [date]
6. Upload Week14_Milestone_Report.pdf to Canvas before the scheduled lab defense; a repository commit does not replace the Canvas submission. [date]
7. During the defense, open the Canvas artifact and the repository; be ready to show make reset-demo, an investigation on CNC-03, the control chart with stepped limits, and the 429 and 422 error paths. [date]

8. Personal reflection (Shritej Gunda)

[Write 150 to 250 words in your own voice. Replace each prompt.]

[Which defect from the table in section 2 would you have shipped without the review, and what in your own testing habit let it through?]

[Explain the SELECT ... FOR UPDATE decision in the overdue check, or the per-subgroup control limits, in your own words as you would in the defense.]

[Describe one AI suggestion from the final wave that you revised or rejected, how you found the problem, and what you learned about MySQL or about testing from it.]

[What did running the whole suite against a real MySQL schema teach you that unit tests alone would not have?]

9. Personal-work summary

[List the artifacts you personally produced or verified for this milestone, one line each, with the file path or screen name: for example the migrations you ran up and down, the queries you checked under EXPLAIN, the README commands you executed, and the screenshots or video you recorded.]